10.12 Lorentz Invariance of Cross Sections

The cross-section quantifies the likelihood of a scattering process (most commonly used with two incoming particles). A property of cross sections is their **Lorentz invariance**: σ is the same in every frame in which the two beams are collinear, and in any other frame it is defined so that it keeps this value. This short note explains why cross sections are Lorentz invariant, despite their dimensional interpretation as "areas," and clarifies a common misconception.

The Paradox of Cross Sections as Invariant "Areas"

Cross sections (σ) have dimensions of area ([Length]^2), leading to the apparent contradiction: *"Areas generally change under boosts; how can σ, defined as an area, be Lorentz invariant?"*
The resolution has two parts.

- **Boosts along the beams.** σ is an area transverse to the beams, and a boost along them does not contract transverse lengths, so it leaves σ unchanged (the note on LecNote10 p. 9). In every frame where the beams are collinear, the flux factor that the definition below gives, $4E_1E_2|\mathbf{v}_1 - \mathbf{v}_2|$ ($2E$ per particle from the normalization, times the relative speed), equals the invariant flux factor below.
- **Other frames.** After a transverse boost the beams are no longer collinear, and $4E_1E_2|\mathbf{v}_1 - \mathbf{v}_2|$ differs from the invariant flux factor. There σ is *defined* by the invariant expression: Srednicki's "convention" (§11, above (11.22)).

With this definition σ is a **Lorentz scalar** built from invariant components:

$$\sigma = \frac{|\mathcal{M}|^2 \cdot (\text{LIPS})}{\text{Flux Factor}}, \tag{10.12.1}$$

where:

- $|\mathcal{M}|^2$: Lorentz-invariant scattering amplitude.
- LIPS: Lorentz-invariant phase space.
- Flux Factor: Lorentz-invariant collision flux.

For two colliding particles with four-momenta p_1, p_2,

$$\text{Flux Factor} = 4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}.$$

The flux factor is manifestly invariant, as it is written in terms of Lorentz invariant quantities. The "tricky" point here is that we motivated the flux

factor from a particular frame and then "dressed it up" with normalization factors from $\langle i|i\rangle/V$ (which, for two particles, scaled as V). The supplement CrossSection_GoldenRule (section Cross Sections) derives it from $2E_a 2E_b(v_a - v_b)$ for head-on beams and gives its center-of-mass and fixed-target forms, $4|\mathbf{p}_a^*|\sqrt{s}$ and $4|\mathbf{p}_a|m_b$.

Definition

Cross sections are defined as:

$$\sigma = \frac{\text{interaction rate per target particle}}{\text{incident flux (particles per unit area per unit time)}}. \tag{10.12.2}$$

A common misconception is that the flux is invariant because the relative velocity is. It is not: the relative velocity v_{rel} is frame-dependent, and the invariant flux is its relativistic *generalization*.

Experimental Consistency (Convenience)

Cross section invariance ensures experimental results are frame-independent:

- **Example**: The same scattering process measured in a collider (center-of-mass frame) and in a fixed-target lab frame, at the same $\sqrt{s}$, yields identical σ.
- **Importance**: the invariant definition gives one σ for all inertial frames, so a prediction made in one frame can be compared with a measurement made in another.